GATEWICK COSTUME & PROP RENTAL HOUSE

Computer Network Design & Implementation Report

Cisco Packet Tracer Case Study

Field	Details
Author	Sanika Kangane
Program	B.Tech CSE '29

Prepared for college submission

1. Introduction

Gatewick Costume & Prop Rental House currently has no computer network. This case study proposes a structured IPv4 network for the Warehouse, Fitting Studio, Administration and IT departments, together with a point-to-point WAN connection to the offsite Prop Workshop. The design uses VLSM with 20% spare capacity, VLAN segmentation, IEEE 802.1Q trunking and Router-on-a-Stick inter-VLAN routing.

2. Objectives

- Design an Extended Star topology.
- Apply VLSM to 10.200.0.0/22 with 20% spare capacity.
- Separate departments using VLANs 10, 20, 30 and 40.
- Use access ports for end devices and trunk ports for uplinks.
- Use IEEE 802.1Q and Router-on-a-Stick for inter-VLAN routing.
- Provide a /30 point-to-point WAN link to the Prop Workshop.
- Verify VLANs, trunks, router interfaces and connectivity.

3. Proposed Topology

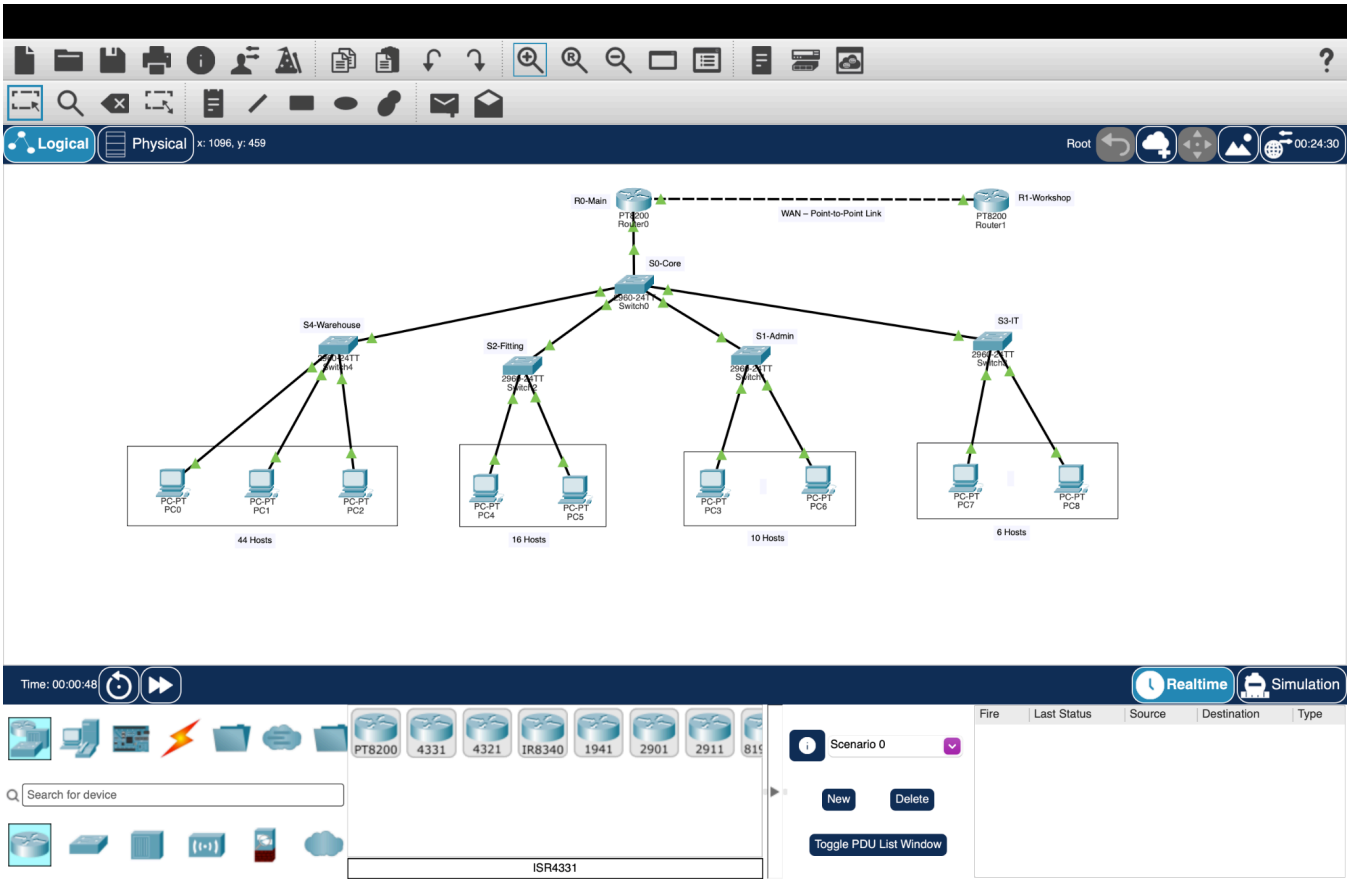


Figure 1. Cisco Packet Tracer topology for the Gatewick network

Source: user-provided Cisco Packet Tracer screenshot. The figure is included as the actual topology reference for this report.

The recommended topology is an Extended Star. S0-Core is the central switch, department switches connect to it, R0-Main provides Router-on-a-Stick routing, and R0-Main connects to R1-Workshop through the point-to-point WAN.

Technology	Purpose
VLANs	Logical separation of departments
Access ports	Connect end devices to their VLAN
Trunk ports	Carry multiple VLANs
IEEE 802.1Q	VLAN tagging on trunks
Router-on-a-Stick	Inter-VLAN routing using subinterfaces
VLSM	Efficient subnet allocation
Point-to-Point WAN	Remote workshop connectivity
IPv4 / ICMP / ARP	Addressing and connectivity testing

4. Physical Connections

From	Interface	To	Interface	Purpose
R0-Main	G0/0/0	S0-Core	Gi0/1	Router-to-core trunk
R0-Main	G0/0/1	R1-Workshop	G0/0/0	WAN / leased line
S0-Core	Fa0/1	S2-Fitting	Gi0/1	Department uplink
S0-Core	Fa0/2	S4-Warehouse	Gi0/1	Department uplink
S0-Core	Gi0/2	S1-Admin	Gi0/1	Department uplink
S0-Core	Fa0/3	S3-IT	Gi0/1	Department uplink

PC Connections

Switch	Connections
S4-Warehouse	Fa0/1 → PC0; Fa0/2 → PC1; Fa0/3 → PC2
S2-Fitting	Fa0/1 → PC4; Fa0/2 → PC5
S1-Admin	Fa0/1 → PC3; Fa0/2 → PC6
S3-IT	Fa0/1 → PC7; Fa0/2 → PC8

5. VLSM and IP Addressing

Starting block: 10.200.0.0/22. The following VLSM plan follows the supplied host requirements and 20% spare-capacity calculations.

Network	Hosts	20% Capacity	Subnet	Mask	Usable Range	Broadcast	Gateway
Warehouse	44	53	10.200.0.0/26	255.255.255.192	10.200.0.1–10.200.0.62	10.200.0.63	10.200.0.1
Fitting Studio	16	20	10.200.0.64/27	255.255.255.224	10.200.0.65–10.200.0.94	10.200.0.95	10.200.0.65
Administration	10	12	10.200.0.96/28	255.255.255.240	10.200.0.97–10.200.0.110	10.200.0.111	10.200.0.97
IT	6	8	10.200.0.112/28	255.255.255.240	10.200.0.113–10.200.0.126	10.200.0.127	10.200.0.113
WAN	2	2	10.200.0.128/30	255.255.255.252	10.200.0.129–10.200.0.130	10.200.0.131	—

Remaining spare address space: 10.200.0.132 – 10.200.3.255.

6. VLAN Design

VLAN ID	Department	Network	Gateway
10	Warehouse	10.200.0.0/26	10.200.0.1
20	Fitting	10.200.0.64/27	10.200.0.65
30	Administration	10.200.0.96/28	10.200.0.97
40	IT	10.200.0.112/28	10.200.0.113

Access ports: S4-Warehouse Fa0/1–Fa0/3 → VLAN 10; S2-Fitting Fa0/1–Fa0/2 → VLAN 20; S1-Admin Fa0/1–Fa0/2 → VLAN 30; S3-IT Fa0/1–Fa0/2 → VLAN 40. Department uplinks are trunks. S0-Core Gi0/1 connects to R0-Main as a trunk.

7. PC IP Addressing

PC	Department	IP Address	Subnet Mask	Gateway
PC0	Warehouse	10.200.0.2	255.255.255.192	10.200.0.1
PC1	Warehouse	10.200.0.3	255.255.255.192	10.200.0.1
PC2	Warehouse	10.200.0.4	255.255.255.192	10.200.0.1
PC4	Fitting	10.200.0.66	255.255.255.224	10.200.0.65
PC5	Fitting	10.200.0.67	255.255.255.224	10.200.0.65
PC3	Administration	10.200.0.98	255.255.255.240	10.200.0.97
PC6	Administration	10.200.0.99	255.255.255.240	10.200.0.97
PC7	IT	10.200.0.114	255.255.255.240	10.200.0.113
PC8	IT	10.200.0.115	255.255.255.240	10.200.0.113

DNS: 0.0.0.0 / blank because no DNS server is configured.

8. Core Networking Concepts

Concept	Simple Explanation
VLAN	A logical network segment that separates devices into different broadcast domains.
Access Port	A switch port assigned to one VLAN, normally used for an end device.
Trunk Port	A link carrying traffic for multiple VLANs.
IEEE 802.1Q	The standard used for VLAN tagging on trunk links.
Subinterface	A logical interface created under a physical router interface for a VLAN.
Router-on-a-Stick	Inter-VLAN routing using one physical router interface with multiple subinterfaces.
Inter-VLAN Routing	Routing traffic between different VLANs or IP subnets.
Default Gateway	The router address used to reach other networks.
VLSM	Variable Length Subnet Masking, which assigns subnet sizes according to host needs.
Subnet Mask	Identifies the network and host portions of an IPv4 address.
WAN	A network connection covering a larger geographic area.
Point-to-Point Connection	A direct link between two devices; /30 provides two usable IPv4 addresses here.
MAC Address	A Layer 2 address used for Ethernet frame delivery.
IP Address vs MAC Address	IP identifies an interface at Layer 3; MAC identifies an Ethernet interface at Layer 2.
ARP	Finds the MAC address associated with a local IPv4 address.
ICMP / Ping	ICMP provides control/diagnostic messages; ping uses ICMP Echo to test reachability.
DHCP	Can automatically provide IP configuration such as address, mask, gateway and DNS.
DNS	Translates names into IP addresses.

9. Router-on-a-Stick Configuration

```
enable
configure terminal

interface gigabitEthernet 0/0/0
no ip address
no shutdown
exit

interface gigabitEthernet 0/0/0.10
encapsulation dot1Q 10
ip address 10.200.0.1 255.255.255.192
exit
```

```
interface gigabitEthernet 0/0/0.20
encapsulation dot1Q 20
ip address 10.200.0.65 255.255.255.224
exit

interface gigabitEthernet 0/0/0.30
encapsulation dot1Q 30
ip address 10.200.0.97 255.255.255.240
exit

interface gigabitEthernet 0/0/0.40
encapsulation dot1Q 40
ip address 10.200.0.113 255.255.255.240
exit

end
```

WAN addressing: R0-Main G0/0/1 = 10.200.0.129/30; R1-Workshop G0/0/0 = 10.200.0.130/30. R0 ↔ R1 ping was successfully verified with 100% success.

10. VLAN Verification

Command Output

```
VLAN Name                              Status    Ports
10   Warehouse                         active    Fa0/1 Fa0/2 Fa0/3
20   Fitting                           active
30   Administration                    active
40   IT                                active
```

Figure 1. Reconstructed Verification Output — S0-Core show vlan brief. Not an actual Packet Tracer screenshot.

11. Trunk Verification

Command Output

```
Port        Mode   Encapsulation   Status        Native vlan
Fa0/1       on     802.1q          trunking      1
Fa0/2       on     802.1q          trunking      1
Fa0/3       on     802.1q          trunking      1
Gi0/1       on     802.1q          trunking      1
Gi0/2       on     802.1q          trunking      1

Vlans allowed and active:
1,10,20,30,40
```

Figure 2. Reconstructed Verification Output — S0-Core show interfaces trunk. Not an actual Packet Tracer screenshot.

12. Router Interface Verification

Command Output

```
Interface              IP-Address      Status     Protocol
G0/0/0                 unassigned      up         up
G0/0/0.10              10.200.0.1      up         up
G0/0/0.20              10.200.0.65     up         up
G0/0/0.30              10.200.0.97     up         up
G0/0/0.40              10.200.0.113    up         up
G0/0/1                 10.200.0.129    up         up
```

Figure 3. Reconstructed Verification Output — R0-Main show ip interface brief. Not an actual Packet Tracer screenshot.

Command Output

```
interface g0/0/0.10
 encapsulation dot1Q 10
 ip address 10.200.0.1 255.255.255.192

interface g0/0/0.20
 encapsulation dot1Q 20
 ip address 10.200.0.65 255.255.255.224

interface g0/0/0.30
 encapsulation dot1Q 30
 ip address 10.200.0.97 255.255.255.240

interface g0/0/0.40
 encapsulation dot1Q 40
 ip address 10.200.0.113 255.255.255.240
```

Figure 4. Reconstructed Verification Output — Router-on-a-Stick configuration. Not an actual Packet Tracer screenshot.

13. Connectivity Testing

Command Output

```
> ping 10.200.0.3
Reply from 10.200.0.3: bytes=32 time<1ms TTL=128
Reply from 10.200.0.3: bytes=32 time<1ms TTL=128
Reply from 10.200.0.3: bytes=32 time<1ms TTL=128
Reply from 10.200.0.3: bytes=32 time<1ms TTL=128

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

Figure 5. Reconstructed Verification Output — PC0 → PC1. Not an actual Packet Tracer screenshot.

Command Output

```
> ping 10.200.0.66
Reply from 10.200.0.66: bytes=32 time<1ms TTL=127
Reply from 10.200.0.66: bytes=32 time<1ms TTL=127
Reply from 10.200.0.66: bytes=32 time<1ms TTL=127
Reply from 10.200.0.66: bytes=32 time<1ms TTL=127

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

Figure 6. Reconstructed Verification Output — PC0 → Fitting Studio. Not an actual Packet Tracer screenshot.



Figure 7. Reconstructed Verification Output — PC0 → R1-Workshop. Not an actual Packet Tracer screenshot.

14. DHCP → ARP → DNS → ICMP

DHCP can provide a host with an IP address, subnet mask, default gateway and DNS server. ARP is then used when the host needs the MAC address associated with a local IPv4 address. DNS translates a domain or host name into an IP address. ICMP is used by tools such as ping to test whether a destination is reachable. For this actual Gatewick project, IP addresses were manually configured, DHCP was not configured, DNS was not configured, and ping/ICMP was used for connectivity testing. Therefore, DHCP and DNS are included as concepts, not as implemented services.

15. OSI / TCP-IP Activity Mapping

Activity	Layer	Reason
Barcode scanner updates inventory dashboard	Application layer	Application-level network service.
Switch forwards a frame using MAC address	Data Link layer	Ethernet frames and MAC addresses.
Faulty cable causes signal loss	Physical layer	Cables and signals operate at the physical layer.
Router chooses the best path	Network layer	Routing uses IP addresses.
Firewall tracks an open inventory-system session	Transport layer	TCP sessions and ports are transport-layer functions.

16. Important Cisco Commands

Command	Simple Explanation
enable	Enters privileged EXEC mode.
configure terminal	Enters global configuration mode.
exit	Moves back one configuration level.
end	Returns to privileged EXEC mode.
interface gigabitEthernet 0/0/0	Selects the physical router interface.
no shutdown	Activates the interface.
no ip address	Removes an IP address from the interface.
interface g0/0/0.10	Selects the VLAN 10 subinterface.
encapsulation dot1Q 10	Associates the subinterface with VLAN 10 using 802.1Q.
ip address	Assigns an IPv4 address and subnet mask.
vlan 10	Creates or selects VLAN 10.
name Warehouse	Names the VLAN Warehouse.
switchport mode access	Makes a switch port an access port.
switchport access vlan 10	Places an access port into VLAN 10.
switchport mode trunk	Configures a switch port as a trunk.
show vlan brief	Displays VLANs, status and assigned access ports.
show interfaces trunk	Displays trunk status and VLAN information.
show ip interface brief	Summarizes interface IP addresses and status.
show mac address-table	Displays learned MAC addresses and ports.
ping	Tests reachability using ICMP Echo messages.

17. Troubleshooting

During testing, one failed ping was caused by entering 100.200.0.3 instead of 10.200.0.3. The corrected command, ping 10.200.0.3, returned 4/4 replies with 0% packet loss. The issue was therefore an incorrect destination IP address rather than a network configuration failure.

18. Verification Summary

Test	Expected Result	Result
S0-Core VLANs	VLANs 10,20,30,40 active	Successful
Core/access uplinks	802.1Q trunking	Successful
R0 subinterfaces	Correct gateway IPs and up/up	Successful
PC0 → PC1	Same-VLAN communication	4/4 replies, 0% loss
PC0 → PC4	Inter-VLAN communication	4/4 replies, 0% loss
PC0 → R1	Remote WAN communication	4/4 replies, 0% loss

19. Conclusion

The Gatewick network design provides a practical structure for a company starting without an existing computer network. VLSM efficiently divides the 10.200.0.0/22 block while preserving spare capacity. VLANs separate departments logically, 802.1Q trunks carry multiple VLANs, Router-on-a-Stick enables inter-VLAN routing, and the /30 WAN subnet supports the two-router point-to-point connection. The supplied verification results show successful VLAN, trunk, interface, same-VLAN, inter-VLAN and WAN connectivity tests.

20. Viva Quick Reference

Question	Short Answer
Why VLSM?	To efficiently use IP addresses by assigning different subnet sizes.
Why VLANs?	To logically separate departments.
Why trunk ports?	To carry multiple VLANs over one link.
Why Router-on-a-Stick?	To route between multiple VLANs using one physical router interface and multiple subinterfaces.
Why /30 for WAN?	It provides exactly two usable IPv4 addresses, suitable for a point-to-point link.
What is the default gateway?	The router address used to reach other networks.
What proves the network works?	Successful ping tests with 0% packet loss, including inter-VLAN and WAN tests.

Device Reference

Type	Names
Routers	R0-Main; R1-Workshop
Switches	S0-Core; S4-Warehouse; S2-Fitting; S1-Admin; S3-IT
Warehouse PCs	PC0; PC1; PC2
Fitting PCs	PC4; PC5
Administration PCs	PC3; PC6
IT PCs	PC7; PC8

21. Cisco Command Prompt Verification

The following pages present Cisco Packet Tracer-style command prompt views used to verify the VLANs, trunk links, router subinterfaces, IP addressing, and connectivity of the Gatewick network. They are formatted as command-prompt verification pages for clear documentation.

21.1 S0-Core - Cisco IOS CLI

S0-Core - Cisco IOS CLI

```
S0-Core> enable
S0-Core# show vlan brief

VLAN Name                              Status    Ports
---- --------------------------------- --------- -------------------------------
10   Warehouse                         active    Fa0/1 Fa0/2 Fa0/3
20   Fitting                           active
30   Administration                    active
40   IT                                active

S0-Core# show interfaces trunk
Port      Mode   Encapsulation   Status       Native vlan
Fa0/1     on     802.1q          trunking     1
Fa0/2     on     802.1q          trunking     1
Fa0/3     on     802.1q          trunking     1
Gi0/1     on     802.1q          trunking     1
Gi0/2     on     802.1q          trunking     1

Vlans allowed and active:
1,10,20,30,40
```

Cisco Packet Tracer-style reconstructed command prompt for report documentation

21.2 R0-Main - Cisco IOS CLI

R0-Main - Cisco IOS CLI

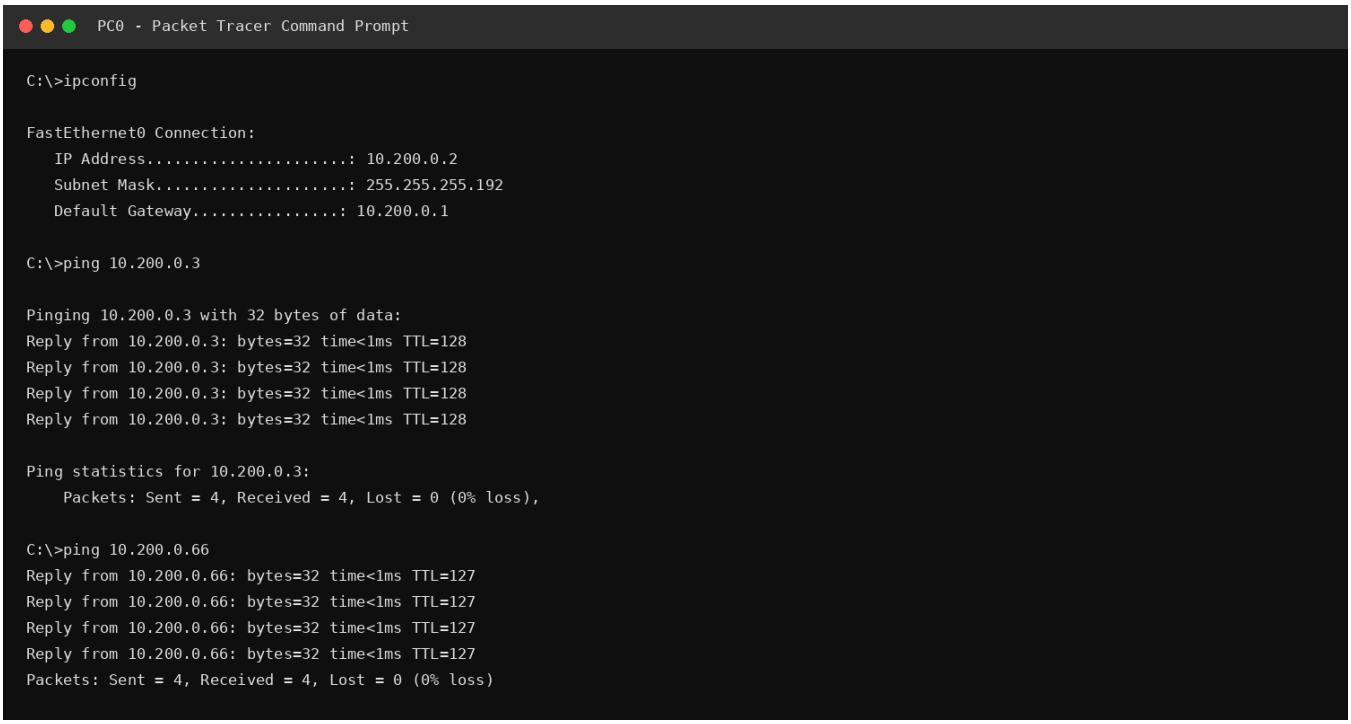
```
R0-Main> enable
R0-Main# show ip interface brief

Interface              IP-Address       Status     Protocol
GigabitEthernet0/0/0   unassigned       up         up
GigabitEthernet0/0/0.10 10.200.0.1      up         up
GigabitEthernet0/0/0.20 10.200.0.65     up         up
GigabitEthernet0/0/0.30 10.200.0.97     up         up
GigabitEthernet0/0/0.40 10.200.0.113    up         up
GigabitEthernet0/0/1   10.200.0.129     up         up

R0-Main# show running-config interface g0/0/0.10
interface GigabitEthernet0/0/0.10
 encapsulation dot1Q 10
 ip address 10.200.0.1 255.255.255.192
R0-Main# show running-config interface g0/0/0.20
interface GigabitEthernet0/0/0.20
 encapsulation dot1Q 20
 ip address 10.200.0.65 255.255.255.224
```

Cisco Packet Tracer-style reconstructed command prompt for report documentation

21.3 PC0 - Packet Tracer Command Prompt



Cisco Packet Tracer-style reconstructed command prompt for report documentation

21.4 R0-Main - WAN Verification

R0-Main - WAN Verification

```
R0-Main> enable
R0-Main# show ip interface brief

GigabitEthernet0/0/1   10.200.0.129     up         up

R0-Main# ping 10.200.0.130

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.200.0.130, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms
```

Cisco Packet Tracer-style reconstructed command prompt for report documentation